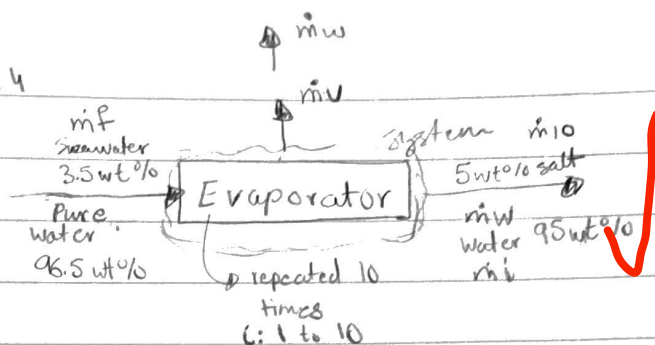


HW #4

1a)



b) Fractional yield of fresh water from the process (kg H₂O recovered / kg H₂O in feed) and the mass fraction w_{us}?

$$0.035 \dot{m}_f = 0.05 \dot{m}_{10}$$

$$\dot{m}_{10} = 0.7 \dot{m}_f$$

$$\dot{m}_f = \dot{m}_w + \dot{m}_{10}$$

$$\dot{m}_f - \dot{m}_{10} = \dot{m}_w$$

$$\dot{m}_w = \dot{m}_f - 0.7 \dot{m}_f = 0.3 \dot{m}_f$$

$$w_{us} = \frac{0.035 \dot{m}_f}{\dot{m}_w}$$

$$f_y = \frac{\dot{m}_w}{0.965 \dot{m}_f}$$

$$0.035 \dot{m}_f = 0.05 \dot{m}_{10}$$

$$\dot{m}_{10} = 0.7 \dot{m}_f$$

$$\dot{m}_f - 0.7 \dot{m}_f = \dot{m}_w$$

$$\dot{m}_w = 0.3 \dot{m}_f$$

$$\dot{m}_w = \dot{m}_f - 0.7 \dot{m}_f = 0.3 \dot{m}_f$$

$$w_{us} = \frac{0.035 \dot{m}_f}{0.3 \dot{m}_f} = 0.1167$$

$$f_y = \frac{0.3 \dot{m}_f}{0.965 \dot{m}_f} = 0.311 \text{ kg H}_2\text{O recovered / kg H}_2\text{O in feed}$$

- No the problem doesn't need a basis b/c the variables cancel out

d) W_{ns}?

$$\dot{m}_n = \dot{m}_f - \frac{n}{10} \dot{m}_w$$

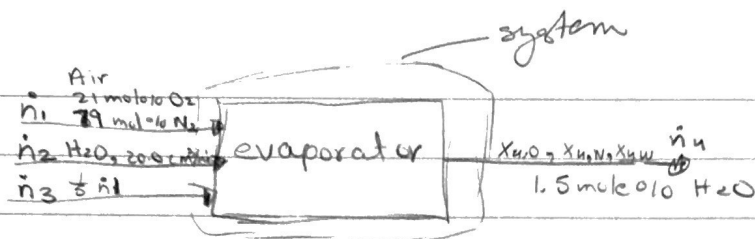
$$\dot{m}_n = \dot{m}_f - \frac{n}{10} (0.3 \dot{m}_f)$$

$$\dot{m}_n = \dot{m}_f (1 - 0.03n)$$

$$W_{ns} = \frac{0.035}{(1 - 0.03n)}$$

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2a



b) DOF

4a.7 5a.1

4b.0 5b.2

5c.1

5d.3

$$7 - 7 = 0 \quad \text{There is a solution} \quad \text{☺}$$

c)

$$\frac{20 \text{ cm}^3}{\text{min}} \times \frac{1 \text{ g}}{\text{cm}^3} \times \frac{\text{mol}}{18.015 \text{ g}} = 1.11 \text{ mol/min} = \dot{n}_2$$

$$X_{H_2O} \cdot \dot{n}_4 = \dot{n}_2$$

$$0.015 \dot{n}_4 = 1.11$$

$$\dot{n}_4 = 74.0 \text{ mol/min}$$

$$\dot{n}_1 + \dot{n}_2 + \dot{n}_3 = \dot{n}_4$$

$$\dot{n}_1 + 1.11 + 0.2\dot{n}_1 = 74$$

$$\frac{1.2\dot{n}_1}{1.2} = \frac{72.89}{1.2}$$

$$\dot{n}_1 = 60.7 \text{ mol/min}$$

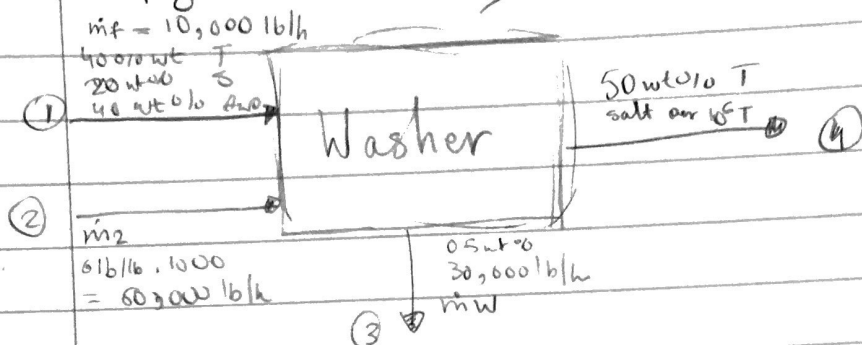
$$\dot{n}_3 = 0.2(60.7) = 12.1 \text{ mol/min} = \dot{n}_3$$

$$N_2: 0.79(\dot{n}_1) = 0.79(60.7) = \frac{47.95}{74} = 0.648 = X_{N_2}$$

$$O_2: 0.21(\dot{n}_1) + \dot{n}_3 = 0.21(60.7) + 12.1 = \frac{24.84}{74} = 0.336 = X_{H_2O}$$

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3. Murphy 2.66



DOF:

4a) 9 5a) 2

4b) 0 5b) 4

5c) 0

5d) 3

$9 - 9 = 0$ completely specified

TiO₂: $0.4 \cdot 10,000 = 4,000 \text{ lb/h}$

Salt: $0.2 \cdot 10,000 = 2,000 \text{ lb/h}$

H₂O w/ pigment: $0.4 \cdot 10,000 = 4,000 \text{ lb/h} + 60,000 \text{ lb/h} = 64,000 \text{ lb/h}$

~~0.5 wt% S~~

TiO₂: $0.4 m_1 = 0.5 m_4$

$$m_4 = \frac{0.4(10,000)}{0.5} = 8,000 \text{ lb/h}$$

$$m_1 + m_2 = m_3 + m_4$$

$$10,000 + 60,000 = m_3 + 8,000$$

$$m_3 = 62,000 \text{ lb/h}$$

$$\text{Salt for } \textcircled{4}: 4000 \left(\frac{182}{100} \right) = 7280 \text{ lb/h}$$

$$0.2(10,000) = S_3(62,000) + 7280$$

$$2000 = 62,000 S_3 + 7280$$

$$S_3 = 0.0322$$

$$S_3 = 3.22\%$$

$$\text{H}_2\text{O}_3 = 96.77 \text{ wt}\%$$

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-4 pts

4) Murphy 2.69

Components for each flow = 25 → 4a

① 3

4b.2

$$2 + 25 = 27$$

② 3

③ 5

④ 3

5a) 1

5b) 4

5c) 2

5d) 20

27

⑤ 2

⑥ 1

⑦ 1

⑧ 1

Mixer: 3

$$27 - 27 = 0$$

Reactor: 5

Completely specified

Splitter: 2

⑨ 2

⑩ 2

Dist 1: 3

⑪ 2

Dist 2: 2

Separator: 5

didn't write out
balances
- 18

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